

Theory of Mechanism and Machine

(Week-7)

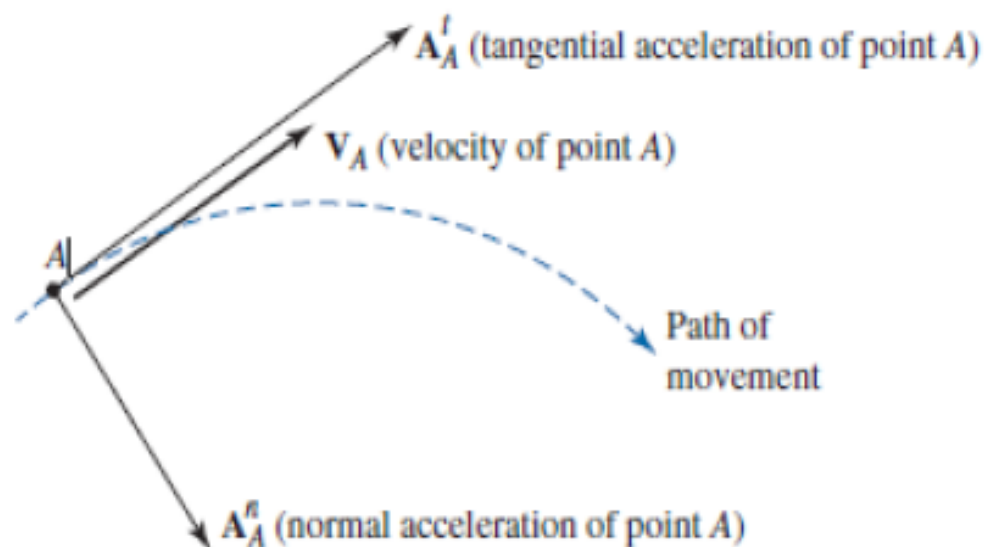
Coriolis Component of Acceleration:

- **Acceleration in Mechanisms ...**

- Acceleration is the rate of change of velocity. It is an important aspects in the design of mechanisms as it directly indicates the inertia forces of the members.

Tangential acceleration a_t

The rate of change of velocity in the tangential direction of the motion is known as tangential acceleration



$$a_t = \frac{dV_A}{dt} = \frac{d(\omega.r)}{dt} = r \frac{d(\omega)}{dt} = \alpha.r$$

Radial or centripetal acceleration a_r :

- The rate of change of velocity towards the centre of rotation:

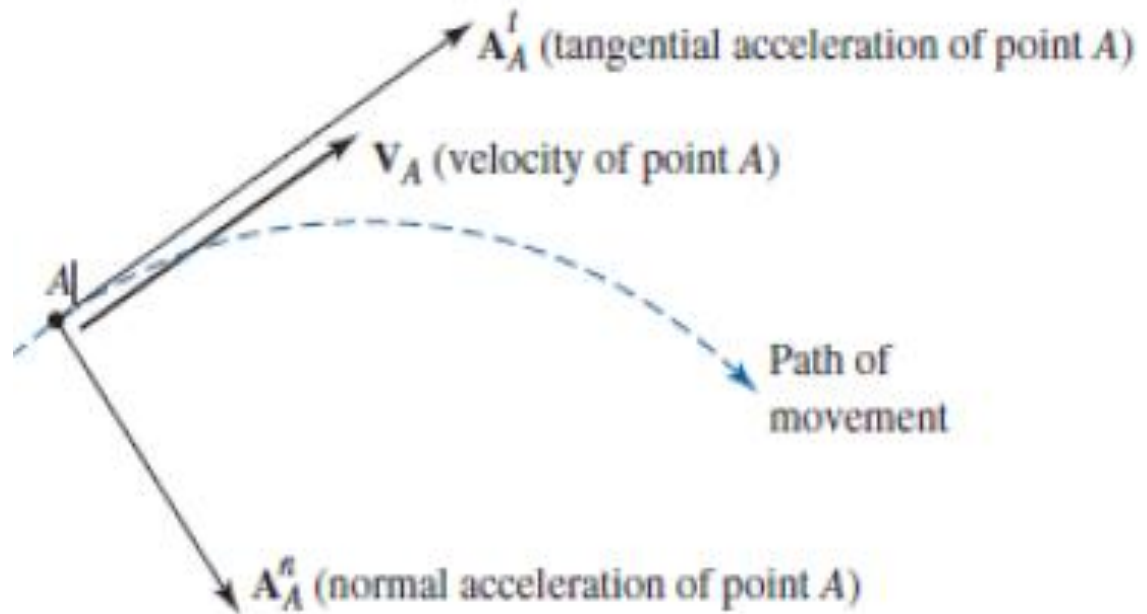
$$a_r = \omega^2 \cdot r = \frac{v_A^2}{r}$$

Total acceleration a_T

Total acceleration is given as:

$$a_T = \vec{a}_r + \vec{a}_t$$

$$a_T = \sqrt{a_r^2 + a_t^2}$$

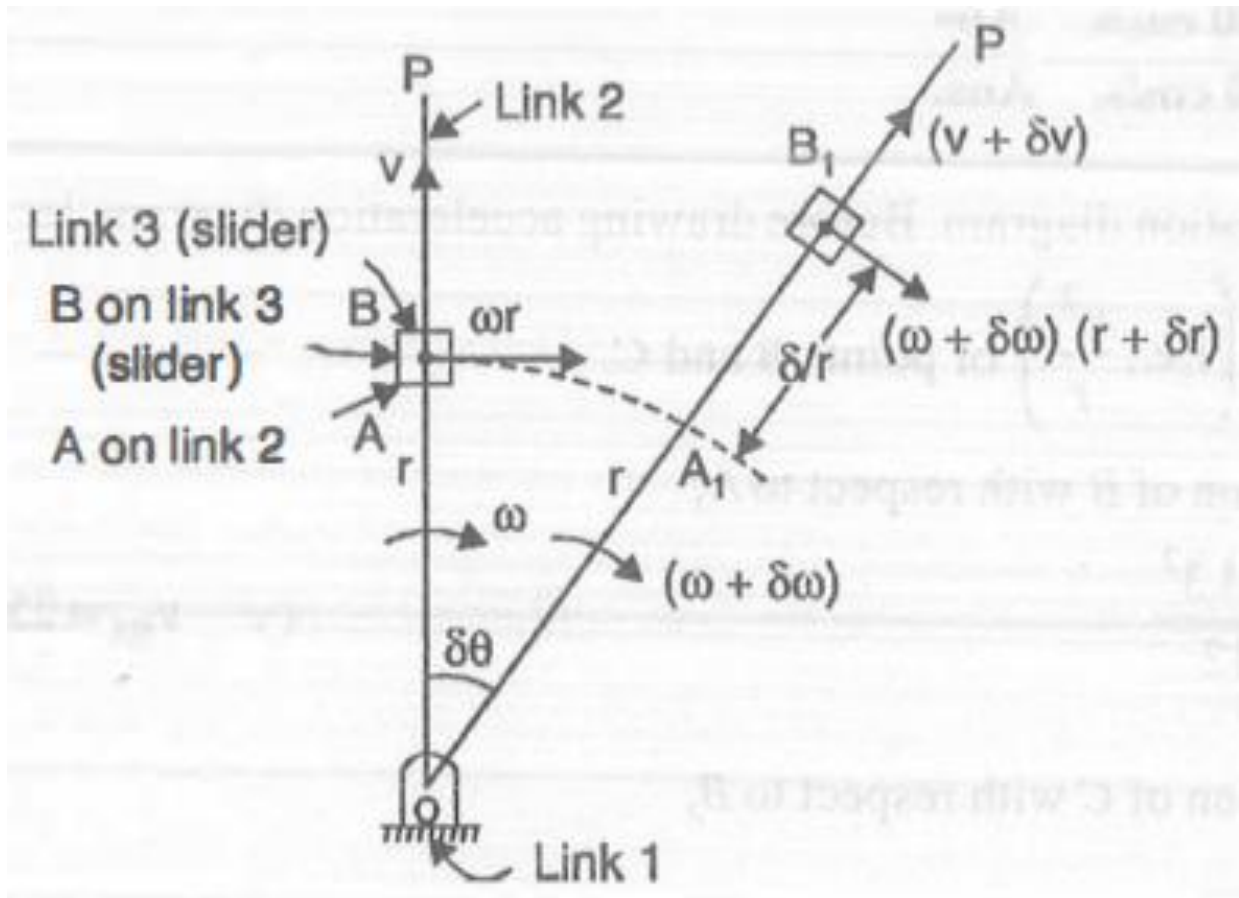


[Coriolis Component of Acceleration (a_c)] ...

When a slider is sliding along a rotating link, Coriolis component of acceleration comes into existence.

In some cases, the point may have its motion relative to a moving body (moving coordinate) system, e.g. motion of a slider on a rotating link.

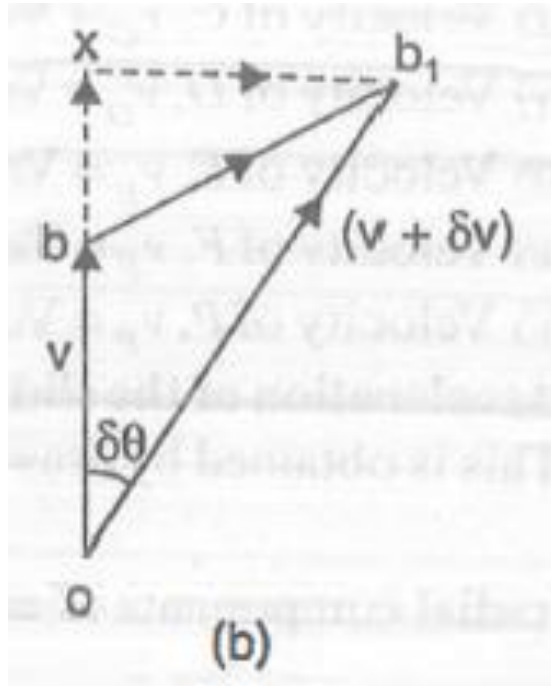
- Consider a link OP rotating clockwise. There is a slider (link 3), sliding along the link OP . Hence the link 3 is moving along a straight line. At any instant ' t ', a point ' A ' on the link OP and the corresponding point on the slider at that instant is B . After a small interval of time ' δt ' the point on the link OP is A_1 and the point on slider is B_1 as shown in Fig.



- Link 1 is fixed, Link 2 is rotating, Link 3 is sliding. Point A is on rotating link and point B is on slider.
- Let ω =Angular velocity of the link OP. This is also the angular velocity of the slider at any time 't'
- v =Linear velocity of the slider B along the link OP at time 't'
- r =Radial distance of point B from O at time t.
- Then tangential velocity of the slider B i.e., velocity perpendicular to OP at time $t = \omega \times r \dots (I)$
- In a short interval of time δt , let $\delta\theta$ be the angular displacement of the link OP and δr , the radial displacement of the slider in the outward direction.
- Let us assume that the link OP is rotating with variable angular velocity and the slider B is also moving with variable linear velocity along the link OP.

- *After the short interval of time δt , let*
- *$(\omega + \delta\omega)$ = Angular velocity of the link OP. This is also the angular velocity of the slider after time δt*
- *$v + \delta v$ = Linear velocity of the slider on the link OP*
- *$r + \delta r$ = Radial distance of the slider.*
- *Then tangential velocity of the slider after time $\delta t = (\omega + \delta\omega)(r + \delta r)$*
- *The slider is rotating side by side it is sliding also. Hence slider is having rotational motion and also linear motion along the link OP.*

Change of velocity along OP and perpendicular to OP due to linear velocity (i.e., sliding velocity).



$$bx = ob_1 \cos \delta\theta - ob$$

$$= ob_1 \cos \delta\theta - ob$$

$$= (v + \delta v) \cos \delta\theta - v$$

$$= (v + \delta v) \times 1 - v$$

$$= \delta v \uparrow$$

$$xb_1 = ob_1 \sin \delta\theta$$

$$= (v + \delta v) \sin \delta\theta$$

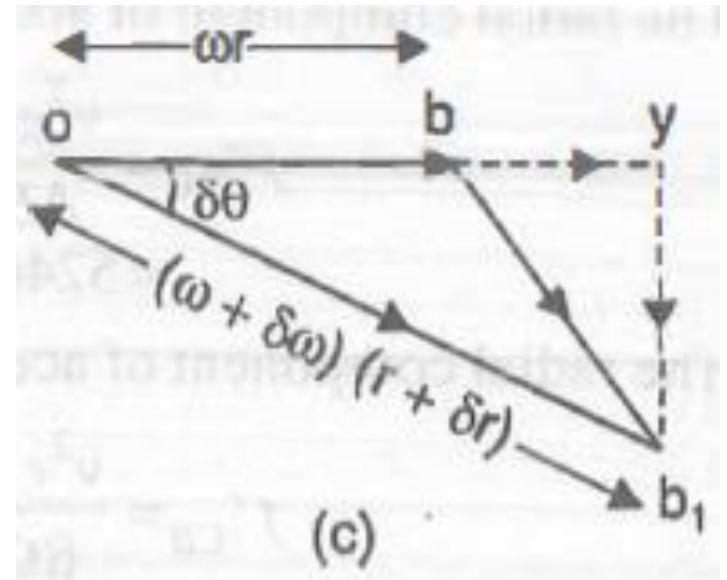
$$= (v + \delta v) \times \delta\theta$$

$$= v \times \delta\theta + \delta v \times \delta\theta$$

$$= v \times \delta\theta \quad (\delta v \times \delta\theta \text{ is negligible,})$$

$$= v \times \delta\theta \rightarrow$$

- The Fig. (c) shows the velocity diagram of the point B (i.e., slider) when its tangential velocities (due to rotation) are considered [i.e., velocities ωr and $(\omega + \delta\omega)(r + \delta r)$].



$$yb_1 = ob_1 \sin \delta\theta \quad \text{where} \quad ob_1 = (\omega + \delta\omega)(r + \delta r)$$

$$= \omega r + \omega \delta r + r \delta \omega + \delta \omega \times \delta r$$

$$= \omega r + \omega \delta r + r \delta \omega \quad (\delta \omega \times \delta r \text{ is negligible})$$

$$= (\omega r + \omega \delta r + r \delta \omega) \sin \delta\theta$$

$$= (\omega r + \omega \delta r + r \delta \omega) \times \delta\theta \quad (\because \sin \delta\theta = \delta\theta \text{ as } \delta\theta \text{ is very small})$$

$$= \omega r \delta\theta + \omega \delta r \delta\theta + r \delta \omega \delta\theta$$

$$= \omega r \delta\theta \quad [\because \delta r \cdot \delta\theta \text{ and } \delta \omega \cdot \delta\theta \text{ are negligible}]$$

$$= \omega r \times \delta\theta \downarrow \quad [\downarrow \text{ shows that change of velocity is in downward direction}] \quad \dots(v)$$

$$\begin{aligned}
 by &= oy - ob \\
 &= ob_1 \cos \delta\theta - ob \\
 &= (\omega r + \omega \delta r + r \delta\omega) \cos \delta\theta - \omega r & (\because ob_1 = \omega r + \omega \delta r + r \delta\omega) \\
 &= (\omega r + \omega \delta r + r \delta\omega) \times 1 - \omega r & (\because \cos \delta\theta = 1 \text{ as } \delta\theta \text{ is very small}) \\
 &= (\omega \delta r + r \delta\omega) & \dots(vi)
 \end{aligned}$$

$$by = (\omega \delta r + r \delta\omega) \rightarrow$$

Total change in velocity of point B in radial direction (*i.e.*, along OP)

$$\begin{aligned}
 &= bx - yb_1 \text{ (} bx \text{ is upward but } yb_1 \text{ is downward)} \\
 &= (\delta v - \omega r \times \delta\theta) & (\because \text{From (iii), } bx = \delta v \text{ and from (v), } yb_1 = \omega r \delta\theta)
 \end{aligned}$$

This total change of velocity of slider B with respect to O is acting radially outwards from O to A .

$\therefore$ Component of acceleration of point B in radial direction (or parallel to OP)

= Rate of total change of velocity in radial direction

$$= \frac{\text{Total change of velocity in radial direction}}{\delta t} = \frac{\delta v - \omega r \delta\theta}{\delta t}$$

$$= \left(\frac{\delta v}{\delta t} - \omega r \frac{\delta\theta}{\delta t} \right) \uparrow$$

In the limit, as $\delta t \rightarrow 0$

Component of acceleration of point B in radial direction (parallel to OP) with respect to O becomes as

$$f^r_{BO} = \left(\frac{dv}{dt} - \omega r \frac{d\theta}{dt} \right) \uparrow$$

$$= f - \omega r \times \omega$$

$$= (f - \omega^2 r) \uparrow$$

f = Linear acceleration of the slider and
 $\omega^2 r$ = Centripetal acceleration.

Similarly the acceleration of point B perpendicular to OP (or in tangential direction) with respect to O

f'_{BO} = Rate of change of total velocity of B perpendicular to OP

$$= \frac{\text{Total change of velocity of } B \text{ perpendicular to } OP}{\delta t}$$

$$= \frac{xb_1 + by}{\delta t} \rightarrow$$

$$= \frac{v\delta\theta + (\omega\delta r + r\delta\omega)}{\delta t} = v \frac{\delta\theta}{\delta t} + \omega \frac{\delta r}{\delta t} + r \frac{\delta\omega}{\delta t} \rightarrow$$

In the limit, as $\delta t \rightarrow 0$, acceleration of point B in tangential direction *i.e.*, perpendicular to OP with respect to O becomes

$$f'_{BO} = v \frac{d\theta}{dt} + \omega \frac{dr}{dt} + r \frac{d\omega}{dt}$$

$$= v \times \omega + \omega \times v + r \times \alpha$$

$$= (2v\omega + r\alpha) \rightarrow$$

$$\left(\because \frac{d\theta}{dt} = \omega, \frac{dr}{dt} = v, \frac{d\omega}{dt} = \alpha \right)$$

The radial component of acceleration of the point A with respect to O,

$$f^r_{AO} = \omega^2 r \downarrow$$

This is acting radially in a direction from C to O.

And the tangential component of acceleration of the point A with respect to O,

$$f^t_{AO} = \alpha \times r \rightarrow$$

This is acting perpendicular to AO and towards right.

Radial and Tangential components of Acceleration of slider B with respect to point A on the link OP.

These components are given as

$$f^r_{BA} = f^r_{BO} - f^r_{AO}$$

$$= (f - \omega^2 r) - (-\omega^2 r)$$

$$= f \uparrow$$

$$\left[\because f^r_{BO} = (f - \omega^2 r) \uparrow \text{ and } f^r_{AO} = \omega^2 r \downarrow \right]$$

(This is acting radially upwards)

$$f^t_{BA} = f^t_{BO} - f^t_{AO}$$

$$= (2v\omega + r\alpha) - \alpha \times r$$

$$= 2v\omega \rightarrow$$

$$\left[\because f^t_{BO} = (2v\omega + r\alpha) \rightarrow \text{ and } f^t_{AO} = r\alpha \rightarrow \right]$$

The above tangential component of acceleration of slider B with respect to the coincident point A on the link is known as **coriolis component of acceleration**.

$2v\omega$ = Coriolis acceleration component

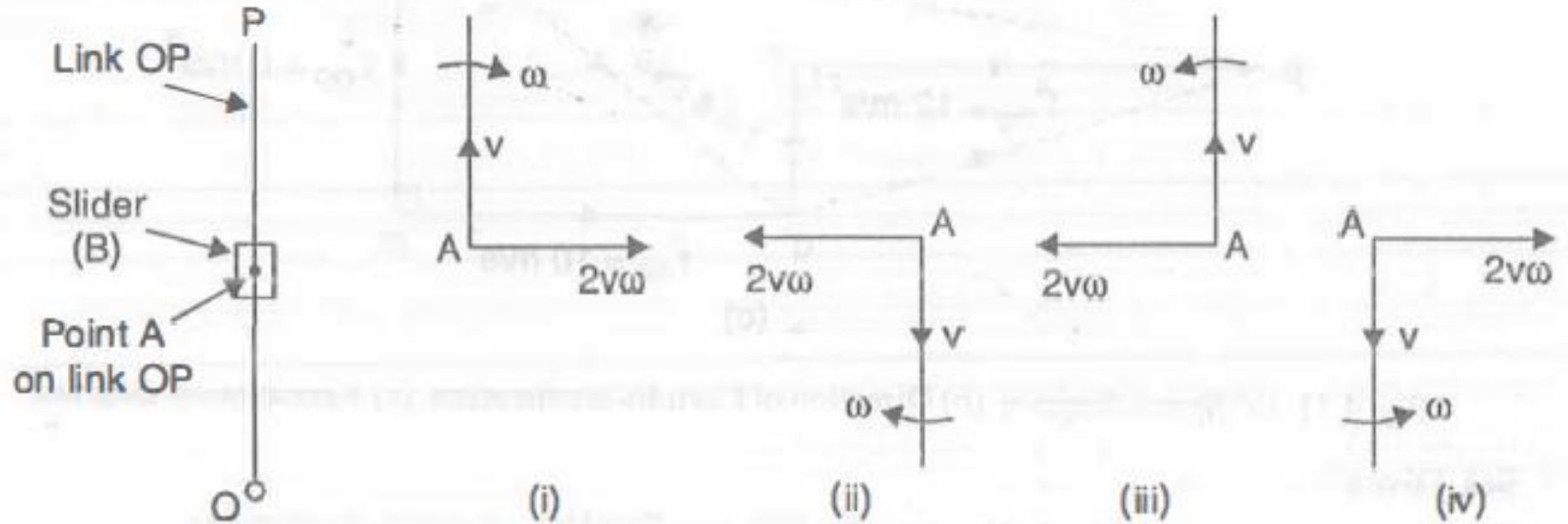
Coriolis acceleration component = $2 \times v \times \omega$

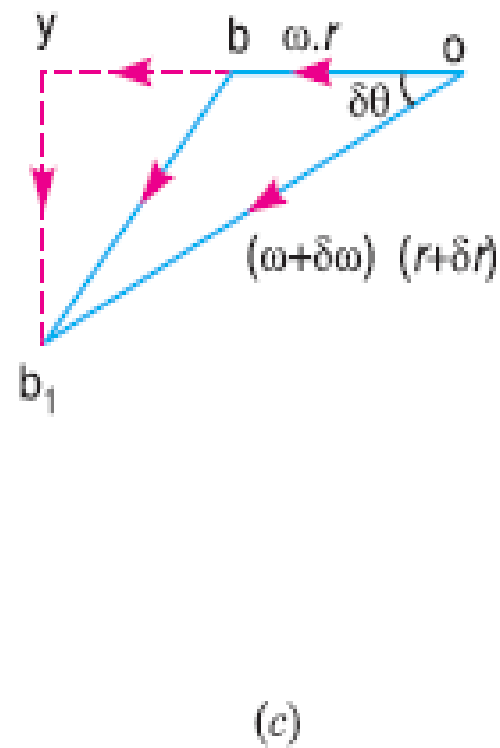
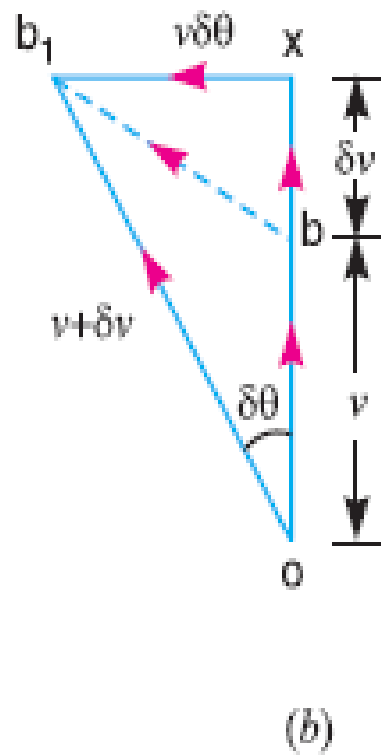
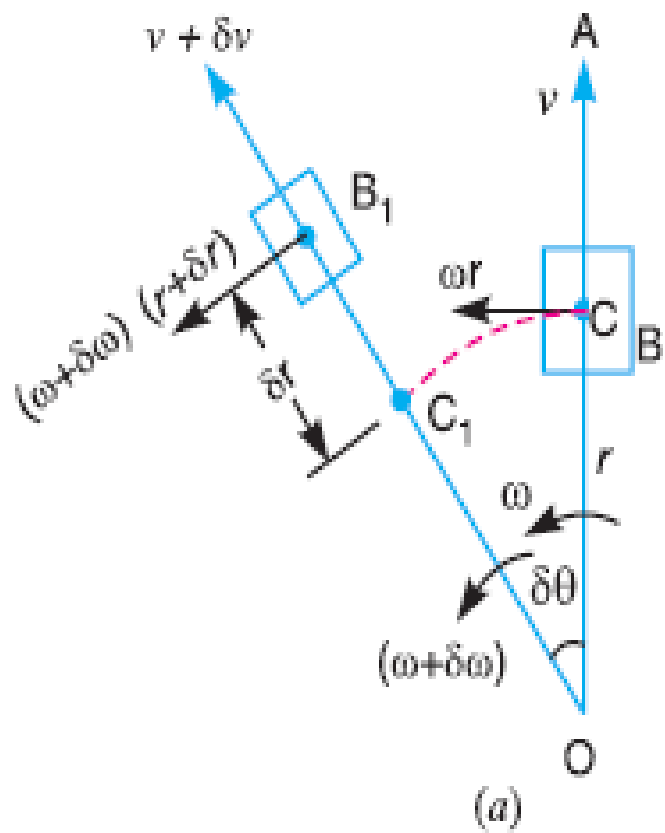
= $2 \times$ Linear velocity of slider $\times$ Angular velocity of rotating link

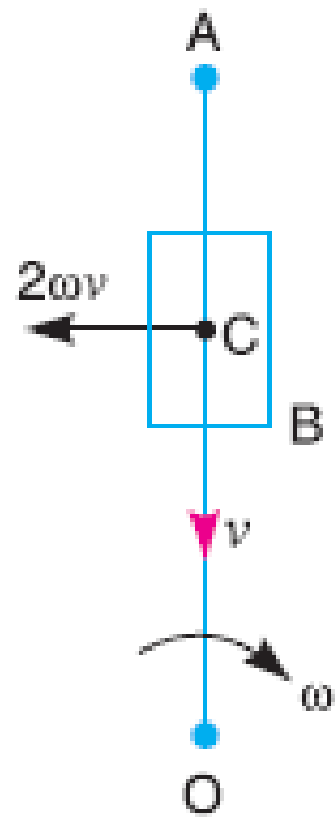
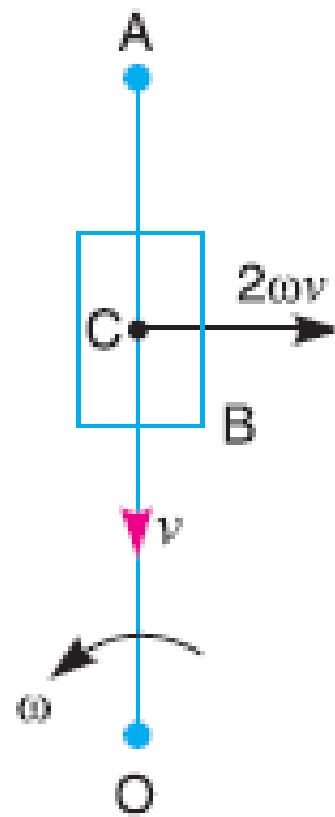
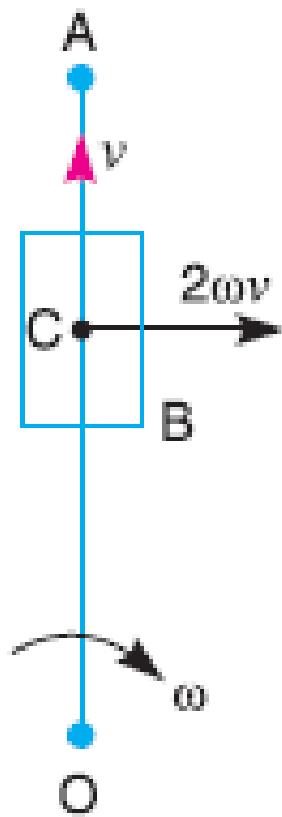
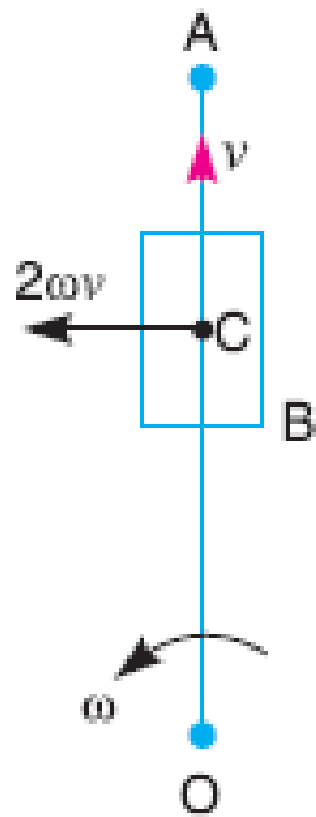
The direction of the Coriolis acceleration component is obtained by rotating the linear velocity vector v through 90° in the direction of rotation of the link.

The direction of ω and v are taken positive if link is rotating clockwise and slider is moving radially outwards. But if the link is rotating anticlockwise, direction of ω is taken -ve. And if slider is moving radially inwards, the direction of v is taken -ve.

The direction of Coriolis component of acceleration for all four possible cases:

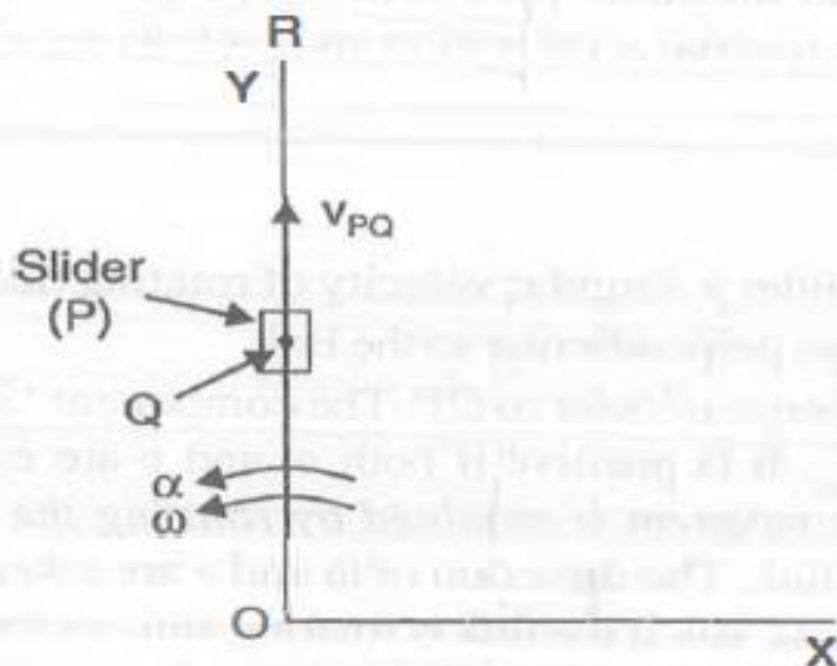






Q 1) A link OR is rotating about point O . At a particular instant when $\theta = 90^\circ$ with x -axis, the angular velocity of the link is 2 rad/s counter-clockwise and angular acceleration is 5 rad/s^2 counterclockwise as shown in Fig. A slider P is sliding along the rod radially outward and at that instant, the distance of the slider from point O is 2 m (i.e., distance $OP = 2 \text{ m}$). The corresponding point on the rod is Q . The velocity and acceleration of the point P at that instant w.r.t. to point Q are 3 m/s and 4 m/s^2 both radially outwards. Find the acceleration of point P relative to point O . Also find

- (i) The acceleration of Q relative to O
- (ii) Acceleration of P relative to Q .



(a)

P and Q are coincident points
Point P is on slider whereas
point Q is on link OR

Sol. Given :

Distance

$$OP = OQ = 2 \text{ m}, \theta = 90^\circ, \alpha = 5 \text{ rad/s}^2 \text{ (counter-clockwise)};$$

$$\omega = 2 \text{ rad/s (counter-clockwise)}; v_{PQ} = 3 \text{ m/s radially outward};$$

Acceleration of P w.r.t. Q $= f = 4 \text{ m/s}^2$ radially outward.

The point Q is fixed to link OR whereas the point P is sliding along OR .

The point Q which is on link has two components of acceleration namely :

(i) Centripetal (or radial) component,

$$\begin{aligned} f'_{QO} &= \omega^2 \times r & (\because r = OP = OQ = 2 \text{ m}) \\ &= 2^2 \times 2 = 8 \text{ m/s}^2. \end{aligned}$$

The centripetal acceleration acts along the link towards centre of rotation. Hence it is acting along QO .

(ii) Tangential component, $f'_{QO} = \alpha \times r = 5 \times 2 = 10 \text{ m/s}^2$.

This is acting perpendicular to QO and in the direction of α .

This is acting perpendicular to QO and in the direction of α .

The point P which is sliding on the link, has also two components of acceleration namely :

(i) The component parallel to OP , $f = 4 \text{ m/s}^2$.

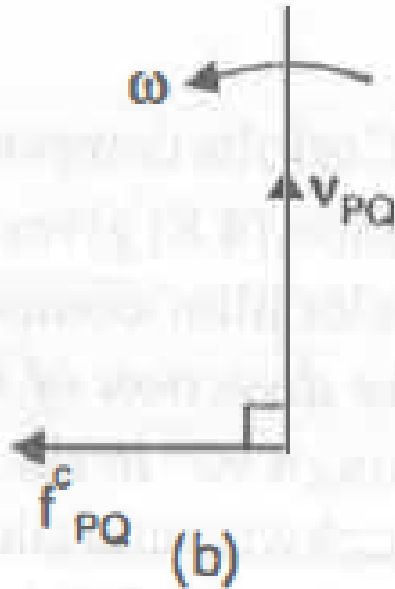
This is also known as linear acceleration of slider.

This component is acting parallel to OP and in the radially outward direction (given).

(ii) Coriolis component of acceleration which is perpendicular to OP . The magnitude of this component is given by

$$f_{PQ}^c = 2 \times \omega \times v_{PQ} = 2 \times 2 \times 3 = 12 \text{ m/s}^2$$

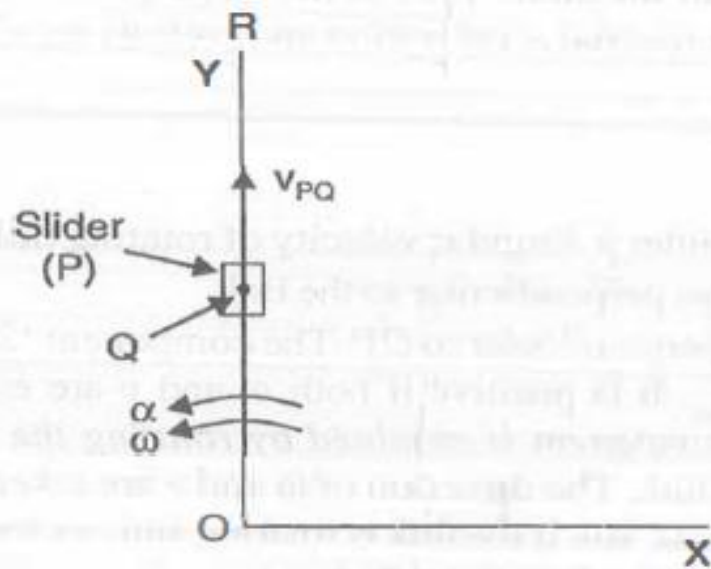
This is acting perpendicular to OP . This direction is obtained by rotating velocity vector v_{PQ} through 90° in the same sense as the sense of rotation of link OP (i.e., in anticlockwise direction) as shown in



2. The acceleration of the **coincident point Q** with respect to O has two components namely centripetal (or radial) and tangential components. The radial component *i.e.*, $f'_{QO} = 8 \text{ m/s}^2$ and acting in the direction of QO . The tangential component *i.e.*, $f'_{QO} = 10 \text{ m/s}^2$ and is acting perpendicular to QO in the direction of α *i.e.*, anticlockwise. These two components are mutually perpendicular. Start the acceleration diagram from the fixed point o' .

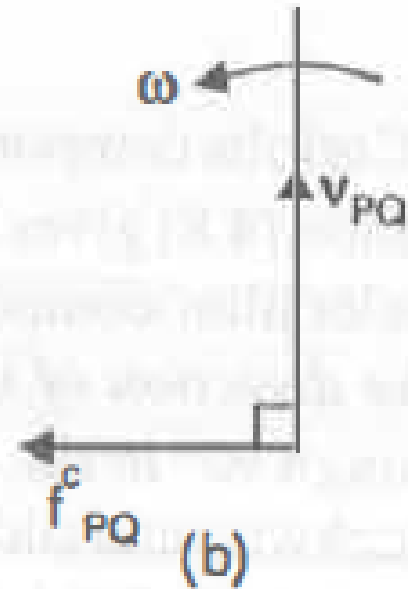
Take any point o' in acceleration diagram as shown in Fig. 4.11 (c). From o' , draw vector $o'm$ parallel to QO in the downward direction. Take this vector $o'm = 8 \text{ m/s}^2$ to some suitable scale. Then this vector represents the radial acceleration of Q with respect to O *i.e.*, f'_{QO} . From point m , draw vector mq' perpendicular to $o'm$ (or perpendicular to OQ in space diagram). This vector should be in the direction of α . This vector represents the tangential acceleration of Q with respect to O *i.e.*, f'_{QO} . Take this vector $mq' = 10 \text{ m/s}^2$. Join q' to o' . The vector $q'o'$ represent the acceleration of Q with respect to O .

3. The acceleration of **point P** (which is sliding along the rod) with respect to coincident point Q has also two components namely sliding acceleration and coriolis component of acceleration. The sliding acceleration *i.e.*, $f = 4 \text{ m/s}^2$ and is acting along the rod OP radially outward. The coriolis component *i.e.*, $f^c_{PQ} = 12 \text{ m/s}^2$ and is acting perpendicular to OP . The direction of this component is obtained by rotating *velocity vector* v_{PQ} through 90° in the same sense as that of the rod OQ rotates. The velocity vector v_{PQ} is acting radially outward

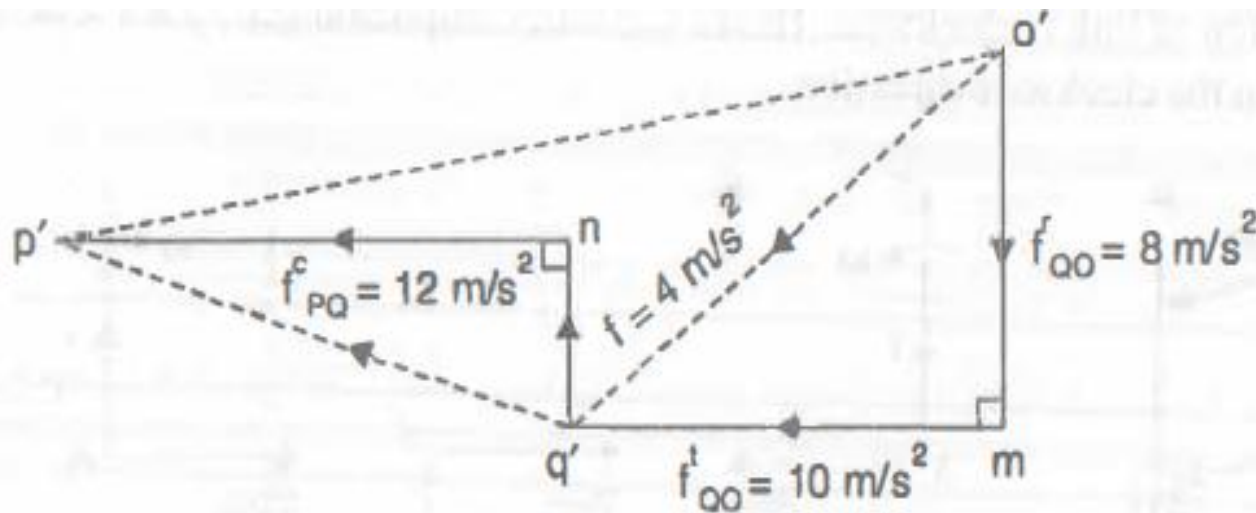


(a)

P and Q are coincident points
Point P is on slider whereas
point Q is on link OR



(b)



From point q' in the acceleration diagram, draw vector $q'n$ parallel to OP in the upward direction from q' . Take $q'n = f = 4 \text{ m/s}^2$. This vector $q'n$ represents the sliding acceleration. Now from point n , draw vector np' perpendicular to $q'n$ (or perpendicular to OP in space diagram) to represent the coriolis acceleration i.e., f^c_{PQ} . Take $np' = 12 \text{ m/s}^2$. The direction of np' should be same as shown in Fig. 4.11 (b). Join point p' to o' . Then $o'p'$ represents the absolute acceleration of P with respect to O .

By measurement :

Vector $o'p' = \text{Acceleration of } P \text{ with respect to } O = 22.4 \text{ m/s}^2. \quad \text{Ans.}$

(i) Acceleration of Q relative to O i.e., $f_{QO} = \text{vector } oq' = 12.8 \text{ m/s}^2. \quad \text{Ans.}$

(ii) Acceleration of P relative to Q i.e., $f_{PQ} = q'p' = 12.65 \text{ m/s}^2. \quad \text{Ans.}$

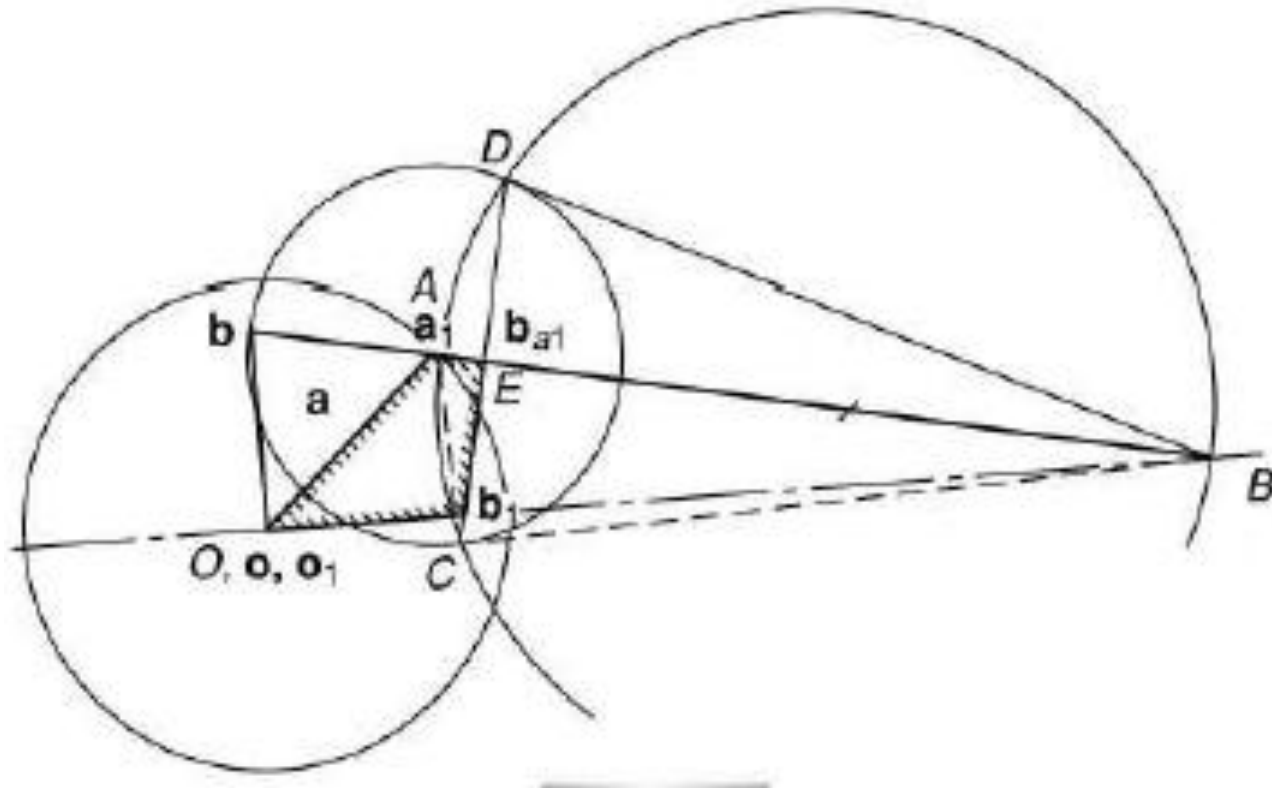
KLEIN'S CONSTRUCTION DIAGRAM:

In this mechanism, the velocity and the acceleration diagrams are made on the configuration diagram itself. The line that represents the crack in the configuration diagram also represents the velocity and the acceleration of its moving end in the velocity and acceleration diagram respectively.

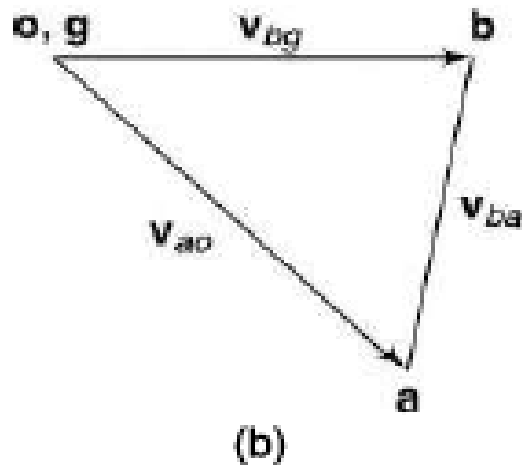
For a slider-crank mechanism, the procedure to make the Klein's construction described below:

Slider Crank Mechanism:

In the below fig, OAB represents a slider crank mechanism.



Its velocity diagram is shown in below.



Velocity Diagram For velocity diagram, let r represent v_{ao} , to some scale. Then for the velocity diagram, length $oa = \omega r = OA$.

From this, the scale for the velocity diagram is known.

Produce BA and draw a line perpendicular to OB through O . The intersection of the two lines

locates the point b . the figure, oab is the velocity diagram which is similar to the velocity diagram of Fig.

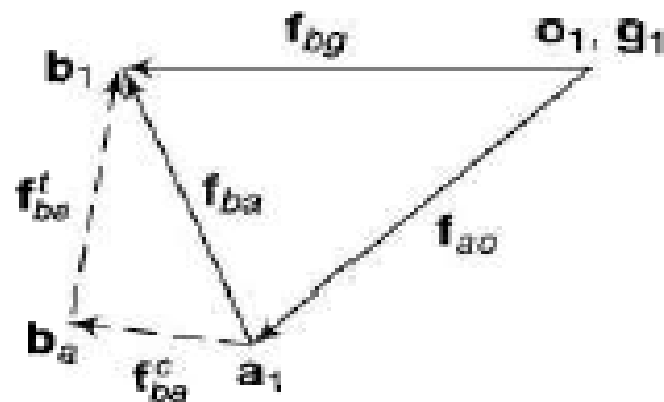
Acceleration Diagram For acceleration diagram, let r represent f_{ao}

$$\therefore \mathbf{o}_1 \mathbf{a}_1 = \omega^2 r = OA$$

This provides the scale for the acceleration diagram.

Make the following construction:

1. Draw a circle with $\mathbf{ab}$ as the radius and $\mathbf{a}$ as the centre.
 2. Draw another circle with AB as diameter.
 3. Join the points of intersections C and D of the two circles. Let it meet OB at $\mathbf{b}_1$ and AB at E .
- Then $\mathbf{o}_1 \mathbf{a}_1 \mathbf{b}_1 \mathbf{b}_1$ is the required acceleration diagram which is similar to the acceleration diagram



(c)

The proof is as follows:

Join AC and BC .

AEC and ABC are two right-angled triangles in which the angle CAB is common. Therefore, the triangles are similar.

$$\frac{AE}{AC} = \frac{AC}{AB} \quad \text{or} \quad AE = \frac{(AC)^2}{AB} \quad \text{or} \quad \mathbf{a_1 b_a} = \frac{(\mathbf{ab})^2}{AB} = f_{ba}^c$$

Thus, this acceleration diagram has all the sides parallel to that of acceleration diagram of Fig. 3.4(c) and also has two sides $\mathbf{o_1a_1}$ and $\mathbf{a_1b_a}$ representing the corresponding magnitudes of the acceleration. Thus, the two diagrams are similar.